A Schreier-Rank Obstruction to Approximate (ℓ_r, ℓ_s) Factorization

Run `fa_banach_001`, lane 6

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Status

This is a conditional reduction packet. The proved part is a necessary condition for the approximate factorization asked about by Mathieu and Tradacete in Remark 6.11 of [1]. The packet does not prove that the needed counterexample operator exists on L_p.

Source Question

Mathieu and Tradacete ask the following in Remark 6.11 of [1]. The crop below is from page 16 of the source PDF.

> then X should contain a subspace isomorphic to ℓ_p, $\ell_{p'}$ or c_0 (although this last option is impossible because of the weak compactness of $L_A R_B$). Hence, the case $X = \ell_s$ as above can also be ruled out by this approach.
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> **Remark 6.11.** We do not know whether every $T \in \mathcal{S}(L_p)$ is approximately (ℓ_r, ℓ_s)-factorizable for some $r < s$. In fact, we do not know whether for an operator $T \in \mathcal{S}(L_p)$, and every $\varepsilon > 0$, there exist $r(\varepsilon) < s(\varepsilon)$, two sequences of finite dimensional spaces $(X_n)_{n\in\mathbb{N}}$, $(Y_n)_{n\in\mathbb{N}}$ and operators
>
> $$\begin{aligned} T_1^\varepsilon :&\ \ L_p \longrightarrow (\oplus X_n)_{\ell_{r(\varepsilon)}} \\ T_2^\varepsilon :&\ \ (\oplus X_n)_{\ell_{r(\varepsilon)}} \longrightarrow (\oplus Y_n)_{\ell_{s(\varepsilon)}} \\ T_3^\varepsilon :&\ \ (\oplus Y_n)_{\ell_{s(\varepsilon)}} \longrightarrow L_p \end{aligned}$$
>
> such that $\|T - T_3^\varepsilon T_2^\varepsilon T_1^\varepsilon\| \leq \varepsilon$. Keeping in mind the previous comment, by Remark 5.3 and the argument in the proof of Theorem 6.9, such a factorization would yield an alternative direct proof of [15, Theorem 2.9].
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> In connection with approximate factorization the following property of strictly

In text form, the first question asks whether every $T \in \mathcal{S}(L_p)$ is approximately (ℓ_r, ℓ_s)-factorizable for some $r < s$. The stronger version asks whether, for every $\varepsilon > 0$, one may factor T within ε through

$$\left(\bigoplus_{n=1}^{\infty} X_n\right)_{\ell_{r(\varepsilon)}} \longrightarrow \left(\bigoplus_{n=1}^{\infty} Y_n\right)_{\ell_{s(\varepsilon)}}$$

for finite-dimensional spaces X_n, Y_n and $r(\varepsilon) < s(\varepsilon)$.

Definitions

Let $\mathcal{S}_1$ denote the first Schreier family:

$$\mathcal{S}_1 = \{F \subset \mathbb{N} : \ F \text{ finite and } |F| \leq \min F\}.$$

An operator $A : E \to F$ is $\mathcal{S}_1$-strictly singular if, for every normalized basic sequence (x_n) in E and every $\eta > 0$, there are $G \in \mathcal{S}_1$ and $0 \neq x \in [x_n]_{n \in G}$ such that

$$\|Ax\| \leq \eta \|x\|.$$

Androulakis, Dodos, Sirotkin, and Troitsky prove the standard properties we use: the class is norm closed and stable under composition with bounded operators, and every operator $\ell_r \to \ell_s$ belongs to $\mathcal{S}_1(\ell_r, \ell_s)$ when $1 < r < s < \infty$ [2].

Proof Intuition

The source factorization has a hidden rank restriction. The middle space changes from an ℓ_r geometry to an ℓ_s geometry with $r < s$. On long disjoint blocks, normalized sums in the domain have size $N^{1/r}$, while their images in the target have size at most on the order of $N^{1/s}$. Since $1/s < 1/r$, long Schreier-admissible averages force the middle map to be $\mathcal{S}_1$-strictly singular. Composition and norm closure then pass this property to every operator obtained by the proposed approximate factorization.

This explains why finite strict singularity is the wrong obstruction: arbitrary maps $\ell_r \to \ell_s$ may fail to be finitely strictly singular, but they still have the first Schreier small-vector property.

Main Lemma

Lemma 1. *Let $1 < r < s < \infty$, let (X_n) and (Y_n) be finite-dimensional Banach spaces, and set*

$$E = \left(\bigoplus_{n=1}^{\infty} X_n \right)_{\ell_r}, \qquad F = \left(\bigoplus_{n=1}^{\infty} Y_n \right)_{\ell_s}.$$

Then every bounded operator $U : E \to F$ is $\mathcal{S}_1$-strictly singular.

Proof. This is the finite-dimensional-decomposition version of the standard gliding-hump proof for operators $\ell_r \to \ell_s$; compare [2, Example 2.4]. Let (x_k) be a normalized basic sequence in E and fix $\eta > 0$. Since E is reflexive and has a finite-dimensional decomposition, the Bessaga-Pelczynski selection principle and the usual sliding-hump reduction allow us, after passing to a subsequence and, if necessary, to a seminormalized difference sequence, to prove the estimate for a normalized block sequence of the FDD. Passing to such a block sequence preserves Schreier admissibility, since the original indices only increase.

Thus suppose (x_k) is a normalized block sequence in E. If $\inf_k \|Ux_k\| = 0$, a singleton Schreier set gives the conclusion. Otherwise pass to a subsequence so that (Ux_k) is seminormalized. A second gliding-hump argument in the finite-dimensional decomposition of F gives a block sequence (y_k) in F and a summable positive sequence (δ_k) such that

$$\|Ux_k - y_k\| \leq \delta_k, \qquad \sup_k \|y_k\| =: M < \infty.$$

For $N \in \mathbb{N}$ put

$$z_N = \sum_{k=N+1}^{2N} x_k.$$

The set $\{N+1, \dots, 2N\}$ belongs to $\mathcal{S}_1$. Since (x_k) is a normalized block sequence in an ℓ_r-sum,

$$\|z_N\| = N^{1/r}.$$

Also, using that (y_k) is a block sequence in an ℓ_s-sum,

$$\|Uz_N\| \le \left\| \sum_{k=N+1}^{2N} y_k \right\| + \sum_{k=N+1}^{2N} \delta_k \le MN^{1/s} + \sum_{k=N+1}^{2N} \delta_k.$$

Therefore

$$\frac{\|Uz_N\|}{\|z_N\|} \le MN^{1/s-1/r} + N^{-1/r} \sum_{k=N+1}^{2N} \delta_k \longrightarrow 0.$$

Choose N large enough that this ratio is at most η. This gives the required Schreier vector. □

Necessary Condition

Theorem 1. *Let X be a Banach space. Suppose $T \in \mathcal{L}(X)$ is a norm limit of operators of the form*

$$X \xrightarrow{A_\varepsilon} \left(\bigoplus_{n=1}^{\infty} X_n^\varepsilon \right)_{\ell_{r_\varepsilon}} \xrightarrow{U_\varepsilon} \left(\bigoplus_{n=1}^{\infty} Y_n^\varepsilon \right)_{\ell_{s_\varepsilon}} \xrightarrow{B_\varepsilon} X,$$

where $1 < r_\varepsilon < s_\varepsilon < \infty$ and all $X_n^\varepsilon, Y_n^\varepsilon$ are finite-dimensional. Then $T \in \mathcal{S}_1(X)$.

Proof. By Lemma 1, each middle operator U_ε is $\mathcal{S}_1$-strictly singular. The ideal property of $\mathcal{S}_1$-strict singularity implies that $B_\varepsilon U_\varepsilon A_\varepsilon$ is $\mathcal{S}_1$-strictly singular on X. Since $\mathcal{S}_1(X)$ is norm closed [2, Proposition 2.2], the norm limit T is also $\mathcal{S}_1$-strictly singular. □

Corollary 1. *Fix $1 < p < \infty$, $p \neq 2$. If there exists an operator*

$$T \in \mathcal{S}(L_p) \setminus \mathcal{S}_1(L_p),$$

then T is not approximately factorizable in the sense of Remark 6.11 of [1], in the reflexive exponent range $1 < r < s < \infty$, even allowing the finite-dimensional-sum formulation.

Proof. The proposed factorization would put T in $\mathcal{S}_1(L_p)$ by Theorem 1, contradicting the choice of T. □

Conditional Dependencies

The proof above is complete as a necessary-condition theorem. To turn the packet into a negative answer to the source question, one still needs the following unproved dependency:

$$\text{construct } T \in \mathcal{S}(L_p) \setminus \mathcal{S}_1(L_p).$$

No such operator was proved in this loop. The packet should therefore remain classified as a conditional reduction.

The exponent range is also part of the scope: the proof uses reflexivity of the ℓ_r-sum and the finite-dimensional-decomposition gliding-hump reduction for $1 < r < s < \infty$. Endpoint variants such as $r = 1$ or $s = \infty$, if intended, require separate verification.

Why Finite Strict Singularity Does Not Work

One might try to replace $\mathcal{S}_1$ by finite strict singularity. That would be too strong. Androulakis-Dodos-Sirotkin-Troitsky record that, for $1 < r < s < \infty$,

$$\mathcal{FSS}(\ell_r, \ell_s) \neq \mathcal{S}(\ell_r, \ell_s) = \mathcal{L}(\ell_r, \ell_s)$$

[2, Example 2.5]. Thus the source definition, whose middle leg is an arbitrary bounded map $\ell_r \to \ell_s$, already permits non-FSS behavior in the middle. The correct necessary obstruction supplied here is the weaker Schreier-rank condition $\mathcal{S}_1$.

Literature and Duplicate Checks

The cheap run indexes were searched for arXiv id `1810.09194`, the title phrase "Strictly singular multiplication operators", and the terms "approximately factorizable", "ell_r ell_s", and "strictly singular multiplication". No direct packet for this source question was found.

Targeted web searches for the exact approximate-factorization phrases found the source paper but no later direct answer. The supporting Schreier-singular facts used here come from arXiv:math/0609039 [2]; this is a tool combination, not a literature answer to Remark 6.11.

Human Review Recommendation

Review the two standard reductions used in Lemma 1: the Bessaga-Pelczynski passage from a basic sequence in an FDD space to a block or difference block sequence, and the gliding-hump approximation of its image by a block sequence in the target FDD. If those are accepted, Theorem 1 follows directly from the closed ideal property of $\mathcal{S}_1$-strict singularity.

References

[1] M. Mathieu and P. Tradacete, *Strictly singular multiplication operators on $\mathcal{L}(X)$*, arXiv:1810.09194, 2018.

[2] G. Androulakis, P. Dodos, G. Sirotkin, and V. G. Troitsky, *Classes of strictly singular operators and their products*, arXiv:math/0609039, 2006.

[3] C. Bessaga and A. Pelczynski, *On bases and unconditional convergence of series in Banach spaces*, Studia Math. 17 (1958), 151–164.